

Total No. of Questions : 8]

SEAT No. :

PB4948

[6263]-621

[Total No. of Pages : 2

**B.E. (Computer Engineering)
INFORMATION AND CYBER SECURITY
(2015 Pattern) (Semester-VIII) (410251)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat, diagrams must be drawn wherever necessary.
- 4) Assume Suitable data if necessary.

- Q1)** a) Define information security. Enlist and explain the elements of information. **[8]**
b) Distinguish between Data Encryption Standard (DES) and Advance Encryption Standard(AES). **[6]**
c) Demonstrate the Diffie-Hellman Key Exchange in detail. **[6]**

OR

- Q2)** a) Define network security. Explain operational model of network security in detail. **[8]**
b) Distinguish between Asymmetric and Symmetric Encryption. **[6]**
c) Solve Diffie-Hellman problem. If $p=7$, $q=17$ select $a=6$, $b=4$. Explain step by step. **[6]**

- Q3)** a) Enlist and explain protocols used in web security. **[8]**
b) Explain the working of SSL session and connection. **[8]**

OR

- Q4)** a) Distinguish between MIME and S/MIME. **[8]**
b) Explain AH and ESP protocol. **[8]**

- Q5)** a) What is computer firewall? Draw and explain Firewall architecture. **[9]**
b) Enlist and explain the limitations and challenges of intrusion detection system. **[8]**

OR

P.T.O.

Q6) a) What are the characteristics and types of firewall? Explain the need of Firewall. [9]

b) What is intrusion detection system and how does it work? [8]

Q7) a) What is cyber stalking? Describe levels of cyber stalking? [9]

b) Explain the concept of PII confidentiality safeguards. [8]

OR

Q8) a) What do you mean by personally identifiable information? Explain the use of PII in terms of confidentiality and cyber forensic. [9]

b) Write a short note on [8]

i) Information Protection Law

ii) Cyber Stalking

